

PENRITH HIGH SCHOOL



MATHEMATICS EXTENSION 1 2013

Preliminary Yearly

Assessor: Mr Ferguson

General Instructions:

- Reading time – 5 minutes
- Working time – **2 hours**
- Write using black or blue pen. Black pen is preferred
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- Show all necessary working in Questions 6 – 9
- Work on this question paper will not be marked.

Total marks – 45

SECTION 1 – Pages 2 -3

5 marks

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section.

SECTION 2 – Pages 4 - 7

40 marks

- Attempt Questions 6 – 9
- Allow about 1 hours 52 minutes for this section.

Section1

Question	Mark
Total	/5

Section 2

Question	Mark
6a	/3
6bc	/7
7a	/2
7bc	/8
8	/10
9	/10
Total	/40

Total	/45
%	

This paper MUST NOT be removed from the examination room

Student Name:

Section I

5 marks

Attempt Questions 1–5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

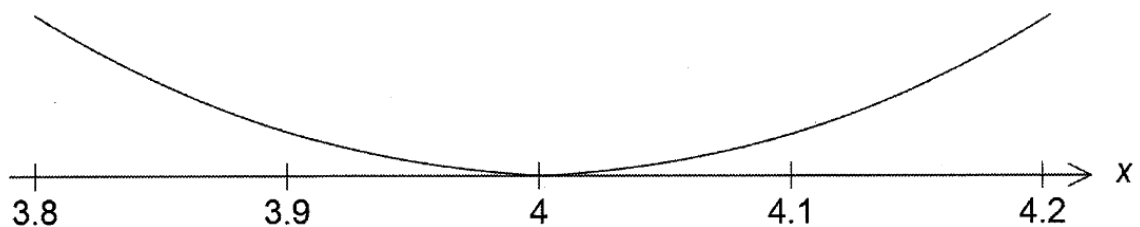
1 What is the solution to the inequality $\frac{x^2 - 4}{2x} > 0$?

- (A) $-2 < x < 0$ or $x > 2$
- (B) $-2 < x < 0$ or $x > 4$
- (C) $-4 < x < 0$ or $x > 2$
- (D) $-4 < x < 0$ or $x > 4$

2 Which of the following is equivalent to the expression $\sin 3\theta$?

- (A) $2\sin \theta - \sin^3 \theta$
- (B) $2\sin \theta - 4\sin^3 \theta$
- (C) $3\sin \theta - \sin^3 \theta$
- (D) $3\sin \theta - 4\sin^3 \theta$

3 Part of the graph of $y = P(x)$, where $P(x)$ is a polynomial of degree 3 is shown below



Which of the following could be the polynomial $P(x)$

- (A) $(x - 4)^3$
- (B) $(x - 5)(x + 4)^2$
- (C) $(x - 1)(x - 4)^2$
- (D) $(x - 1)(x + 2)(x - 4)$

4 The point $P(2,5)$ lies on the graph of the polynomial $y = P(x)$. This polynomial is an odd function. The remainder when $P(x)$ is divided by $(x+2)$ is:

- (A) -5
- (B) 0
- (C) 5
- (D) 2

5 Which of the following is true for the polynomial $P(x) = x^3 - x$

- (A) It has a point of inflexion at $x = 0$
- (B) It has a minimum at $x = \frac{1}{3}$
- (C) It has only one turning point.
- (D) It has a stationary point at $x = -\frac{1}{3}$

Section II

40 marks

Attempt Questions 6–9

Allow about 1 hours and 52 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 6–9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks) Use a SEPARATE writing booklet.

- (a) Solve the inequality $\frac{6}{x^2} \leq \frac{x-5}{x}$ 3
- (b) A and B are the points $(-5, 12)$ and $(4, 9)$ respectively, P is the point which divides AB externally in the ratio $5:2$. Find the co-ordinates of P . 2
- (c) (i) Show that the acute angle between the lines $mx - y = 0$ and $x - my = 0$ can be given by $\tan \theta = \left| \frac{m^2 - 1}{2m} \right|$ where θ is the angle between the lines. 2
- (ii) Find the values of the real number m such that the angle θ is 45° . 3

Question 7 (10 marks) Use a SEPARATE writing booklet.

- (a) If $\cos A = \frac{7}{9}$ and $\sin B = \frac{1}{3}$ where $0 \leq A \leq 90^\circ$, and $0 \leq B \leq 90^\circ$. **4**
- (i) Find the value of $\tan(A + B)$ in simplest surd form.
- (ii) Show that $A = 2B$
-
- (b) (i) Express the polynomial $P(x) = 2x^3 - x^2 - 13x - 6$ as the product of three linear factors. **2**
- (ii) Hence solve the inequality $2x^3 - x^2 - 13x - 6 \leq 0$ **1**
-
- (d) (i) Show that $x^3 - 3x + 1 = 0$ has a root between $x = 1$ and $x = 2$ **1**
- (ii) Using $x = 1.5$ as a first approximation, obtain a better approximation of the root using Newton's Method once. [Answer to 2 decimal places] **2**

Question 8 (10 marks) Use a SEPARATE writing booklet.

- (a) Expand the expression $(2x + 5)^5$ 2
- (b) For what value of n is ${}^nC_5 = {}^nC_9$ 1
- (c) Find, as a rational number, the coefficient of x in the expansion of $\left(x^2 + \frac{1}{2x}\right)^8$ 2
- (d) Suppose $(5 + 2x)^{15} = \sum_{k=0}^{15} t_k x^k$
- (i) Use the binomial theorem to write an expression for t_k , $0 \leq k \leq 15$ 1
- (ii) Show that $\frac{t_{k+1}}{t_k} = \frac{2(15-k)}{5(k+1)}$ 2
- (iii) Hence or otherwise find the largest coefficient of t_k . 2
You may leave your answer in the form ${}^{15}C_k 5^c 2^d$

Question 9 (10 marks) Use a SEPARATE writing booklet.

- (a) Consider the curve given by $y = 1 + 3x - x^3$, for $-2 \leq x \leq 3$
- (i) Find the stationary points and determine their nature. 2
 - (ii) Determine if there exists any points of inflexion. 1
 - (iii) Sketch the curve for $-2 \leq x \leq 3$. 2
 - (iv) What is the minimum value of y for $-2 \leq x \leq 3$? 1
- (b) The sum of the height h of a cylinder and the circumference of its base is 20 metres.
- (i) Show that the volume of the cylinder is $V = \pi r^2(20 - 2\pi r)$, 2
where r is the radius of the cylinder r .
 - (ii) Hence find the maximum volume of the cylinder 2

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

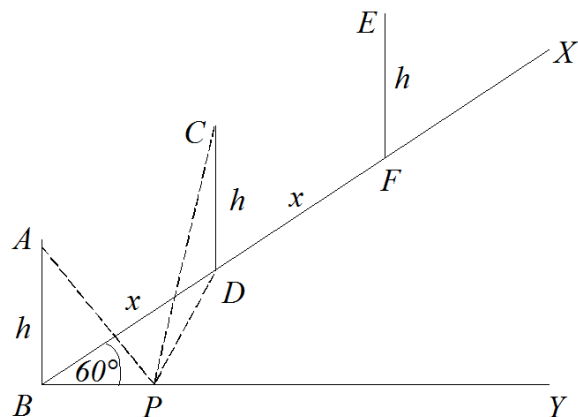
$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

Note: $\ln x = \log_e x$, $x > 0$



- (a) In the above diagram, BX and BY represent two roads intersecting at an angle of 60° . On the road BX are situated three telegraph poles AB , CD and EF , all of equal height, the same distance, x metres apart (i.e $BD = DF = x$). P is a point on the road BY and the angles of elevation of A and C from P are 45° and 30° respectively.
- Show that $BP = h$ and $DP = h\sqrt{3}$
 - By the use of the Sine rule in triangle BDP , show that $BDP = 30^\circ$ and hence that triangle BDP is right angles at P .
 - Prove that $x = 2h$
 - By the use of the cosine rule in triangle PDF , show that $PF = h\sqrt{13}$ and hence show that the angle of elevation of E from P is approximately 15.5° .

- (e) A monic cubic polynomial when divided by $(x^2 + 4)$ leaves a remainder of $x + 8$ and when divided by x leaves a remainder of -4 . Find the polynomial in the form $ax^3 + bx^2 + cx + d$

6 Consider the polynomial $P(x) = 3x^3 + 3x + a$.
If $x - 2$ is a factor of $P(x)$, what is the value of a ?

- 30
- 18
- 18
- 30

7 Let α , β and γ be the roots of $4x^3 - 2x^2 + 3x - 2 = 0$.

What is the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$?

(A) $-\frac{3}{2}$

(B) $-\frac{2}{3}$

(C) $\frac{2}{3}$

8 What are the coordinates of the point that divides the interval joining the points $A(7,1)$ and $B(0,-6)$ internally in the ratio 4:3?

(A) $(3,-3)$

(B) $(3,-2)$

(C) $(4,-2)$

9 Which of the following is equivalent to the expression $\sqrt{\frac{4+4\cos 2x}{1-\cos 2x}}$?

(A) $2\cot^2 x$

(B) $2\cot x$

(C) $2\tan x$

10 What is the acute angle to the nearest degree between the lines $y = 2x - 1$ and $x - 3y + 6 = 0$?

(A) 45°

(B) 54°

(C) 79°

(D) 82°

(d) and show that if Q is the point $(0, 2)$,
then the triangle APQ is both right-angled and isosceles.

(e) Simplify $\frac{4^n - 1}{4^n - 2^n}$ 2

Solve for θ in the range $0 \leq \theta \leq 360^\circ$: $6\cot^2 \theta - 4\cos^2 \theta = 1$ 3

(e) In the binomial expansion $\left(x^2 + \frac{a}{x}\right)^6$ the term independent of x is 240. 3

Find the value of a .

(f) Show that $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$ **2**

(g) Given $y = x^3 - x^2 + x$ show that $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2x(3x-5) = 0$ **2**

(h) Show that if $y = -3x^2 + 1$ then $\frac{dy}{dx} = -6x$ through differentiation by **3**
first principles

(i)

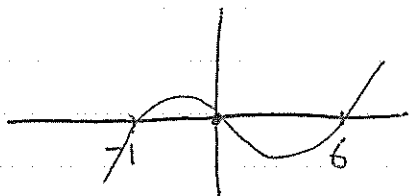
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Multiple choice.

- 1) A
- 2) D
- 3) C
- 4) A
- 5) A

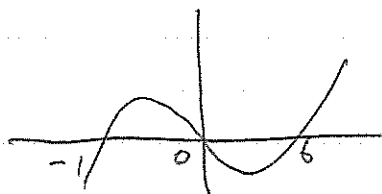
6) a) $\frac{6}{x^2} \leq \frac{x-5}{x}$

For $x > 0$ $6x \leq x^3 - 5x^2$
 $0 \leq x^3 - 5x^2 - 6x$
 $0 \leq x(x^2 - 5x - 6)$
 $0 \leq x(x-6)(x+1)$



$\therefore -1 \leq x \leq 0$ $x \geq 6$

For $x < 0$ $6x \geq x^3 - 5x^2$
 $0 \geq x^3 - 5x^2 - 6x$
 $0 \geq x(x-6)(x+1)$



$x \leq -1$ $0 \leq x \leq 6$

$\therefore x \leq -1$

$\therefore x \geq 6$ or $x \leq -1$

b) $m:n = 5:-2$

$x = \frac{-2(-5) + 5(4)}{5-2}$

$= \frac{10+20}{3}$

$= 10$

$y = \frac{-2(12) + 5(9)}{5-2}$

$= \frac{-24+45}{3}$

$= 7$

$P(10, 7)$

C(i)

$mx - y = 0$

$-y = -mx$

$y = mx$

$x - my = 0$

$-my = -x$

$y = \frac{x}{m}$

$\tan \theta = \left| \frac{m - \frac{1}{m}}{1 + m \cdot \frac{1}{m}} \right|$

$= \left| \frac{\frac{m^2 - 1}{m}}{2} \right|$

$= \left| \frac{m^2 - 1}{m} \times \frac{1}{2} \right|$

$= \left| \frac{m^2 - 1}{2m} \right|$

(ii) $\frac{m^2 - 1}{2m} = 1$ and $\frac{m^2 - 1}{2m} = -1$

$m^2 - 1 = 2m$

$m^2 - 2m - 1 = 0$

$\Delta = (-2)^2 - 4 \times 1 \times -1$

$= 8$

$m = \frac{2 \pm \sqrt{8}}{2}$

$= 1 \pm \sqrt{2}$

$m^2 - 1 = -2m$

$m^2 + 2m - 1 = 0$

$\Delta = (2)^2 - 4 \times 1 \times -1$

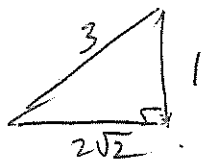
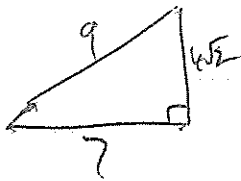
$= 8$

$m = \frac{-2 \pm \sqrt{8}}{2}$

$= -1 \pm \sqrt{2}$

7a)

(i)



$$\begin{aligned}\therefore \sin 2B &= 2 \sin B \cos B \\ &= 2 \cdot \frac{1}{3} \cdot \frac{2\sqrt{2}}{3} \\ &= \frac{4\sqrt{2}}{9}\end{aligned}$$

$$\sin A = \frac{4\sqrt{2}}{9}$$

$$\therefore A = 2B$$

$$(ii) \tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\frac{4\sqrt{2}}{7} + \frac{1}{2\sqrt{2}}}{1 - \frac{4\sqrt{2}}{7} \times \frac{1}{2\sqrt{2}}}$$

$$= \frac{16+7}{14\sqrt{2}} = \frac{5}{7}$$

$$= \frac{23}{14\sqrt{2}} \times \frac{7}{5}$$

$$= \frac{23}{10\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}$$

$$= \frac{23\sqrt{2}}{20}$$

$$b) P(x) = 2x^3 - x^2 - 13x - 6$$

$$P(1) = 2 - 1 - 13 - 6$$

$$P(-1) = -2 - 1 + 13 - 6$$

$$P(2) = 16 - 4 - 26 - 6$$

$$P(-2) = -16 - 4 + 26 - 6 = 0$$

$\therefore (x+2)$ is factor ✓

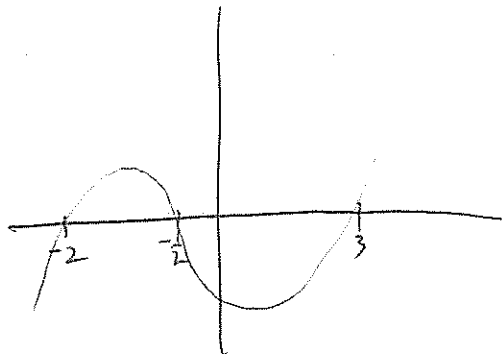
$$\begin{array}{r} 2x^2 - 5x - 3 \\ x+2 \overline{) 2x^3 - x^2 - 13x - 6} \\ \underline{2x^3 + 4x^2} \\ -5x^2 - 13x \\ \underline{-5x^2 - 10x} \\ -3x - 6 \\ \underline{-3x - 6} \\ 0 \end{array}$$

$$(x+2)(2x^2 - 5x - 3)$$

$$(x+2) \frac{(2x-6)(2x+1)}{2}$$

$$\therefore (x+2)(x-3)(2x+1) = P(x)$$

$$(ii) 2x^3 - x^2 - 13x - 6 \leq 0$$



$$x \leq -2 \quad \text{and} \quad -\frac{1}{2} \leq x \leq 3$$

$$d(i) P(1) = 1^3 - 3(1) + 1 = 1 - 3 + 1 = -2$$

$$P(2) = 2^3 - 3(2) + 1 = 8 - 6 + 1 = 3$$

$\therefore$ curve $y = x^2 - 3x + 1$ cuts the x axis between -2 and 3. ✓

$$(ii) a_2 = a_1 - \frac{f(a_1)}{f'(a_1)} \quad f'(x) = 3x^2 - 3$$

$$f(1.5) = (1.5)^3 - 3(1.5) + 1 = -0.125$$

$$f'(1.5) = 3(1.5)^2 - 3 = 3.75$$

$$a_2 = 1.5 - \frac{-0.125}{3.75}$$

$$\approx 1.53$$

$$8a) (2x+5)^5$$

$$= \binom{5}{0}(2x)^0 5^5 + \binom{5}{1}(2x)^1 5^4 + \binom{5}{2}(2x)^2 5^3 + \binom{5}{3}(2x)^3 5^2 + \binom{5}{4}(2x)^4 5^1 + \binom{5}{5}(2x)^5 5^0$$

$$= 3125 + 6250x + 5000x^2 + 2000x^3 + 400x^4 + 32x^5$$

$$(b) {}^{14}C_5 = {}^{14}C_9 \text{ through symmetry}$$

$$(c) T_{k+1} = {}^8C_k (x^2)^{8-k} \left(\frac{1}{2x}\right)^k$$

$$= \frac{{}^8C_k}{2^k} \cdot \frac{x^{16-2k}}{x^k}$$

$$= \frac{{}^8C_k}{2^k} \cdot x^{16-3k}$$

$$\therefore 16-3k=1$$

$$k=5$$

$$\therefore T_6 = \frac{{}^8C_5}{2^5} \cdot x$$

$$= \frac{56}{32}$$

$$= \frac{7}{4}$$

$$d) (5+2x)^{15} = \sum_{k=0}^{15} t_k x^k$$

$$T_{k+1} = {}^{15}C_k 5^{15-k} (2x)^k$$

$$\therefore t_k = {}^{15}C_k 5^{15-k} \cdot 2^k \cdot x^k$$

$$(ii) \frac{t_{k+1}}{t_k} = \frac{{}^{15}C_{k+1} 5^{15-(k+1)} 2^{k+1}}{{}^{15}C_k 5^{15-k} 2^k}$$

$$= \frac{15!}{[15-(k+1)]!(k+1)!} \cdot \frac{5^{15-k} 2^{k+1}}{5^{15-k} 2^k}$$

$$= \frac{15!}{(15-k)!(k+1)!} \cdot \frac{5^{15-k} 2^{k+1}}{5^{15-k} 2^k}$$

$$= \frac{2(15-k)}{5(k+1)}$$

$$\frac{t_{k+1}}{t_k} > 1 \quad \frac{2(15-k)}{5(k+1)} > 1$$

$$2(15-k) > 5(k+1)$$

$$30-2k > 5k+5$$

$$25 > 7k$$

$$3\frac{4}{7} > k$$

$$\therefore k=3$$

$$\therefore t_4 > t_3$$

$$t_4 = {}^{15}C_4 5^{11} 2^4$$

9a) $y = 1 + 3x - x^3$

$-2 \leq x \leq 3$

(i) $\frac{dy}{dx} = 3 - 3x^2$

$= 3(1 - x^2)$

$\frac{d^2y}{dx^2} = -6x$

$\frac{dy}{dx} = 0 \quad 0 = 3(1 - x^2)$

$= (1+x)(1-x)$

$\therefore x = \pm 1$

for $x = 1 \quad y'' = -6 \times 1 < 0 \therefore \text{max}$

$x = -1 \quad y'' = -6 \times -1 > 0 \therefore \text{min}$

$\therefore (1, 3)$ is max

$(-1, 3)$ is min

must do needs to be determined

(ii) $\frac{d^2y}{dx^2} = 0 \quad \therefore -6x = 0$
 $x = 0$

test

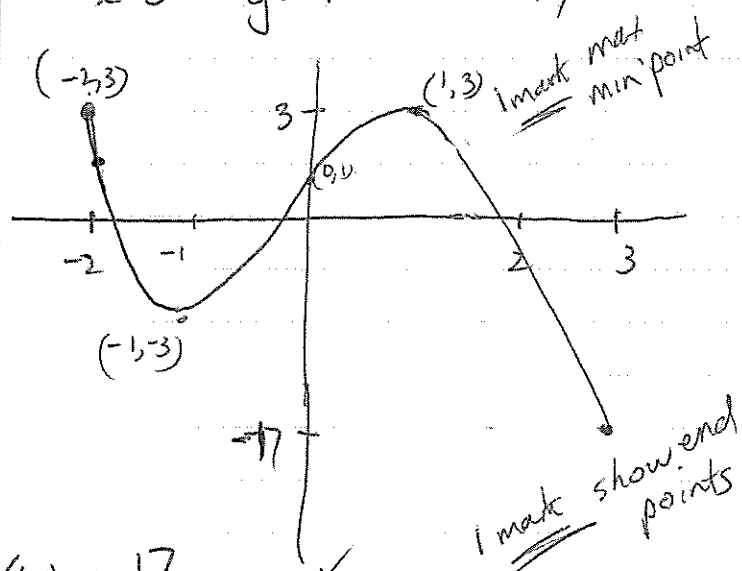
x	$-\frac{1}{2}$	0	$\frac{1}{2}$
y''	3	0	-3

change in concavity

$\therefore (0, 1)$ is inflexion

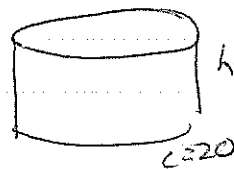
(ii) $x = -2 \quad y = 1 - 6 + 8 = 3$

$x = 3 \quad y = 1 + 9 - 27 = -17$



(iv) -17

b)



$V = \pi r^2 h$

$= \pi r^2 (20 - 2\pi r)$

$= 20\pi r^2 - 2\pi^2 r^3$

$h + c = 20$

$h + 2\pi r = 20$

$h = 20 - 2\pi r$

(ii) $\frac{dV}{dr} = 40\pi r - 6\pi^2 r^2$

$0 = 40\pi r - 6\pi^2 r^2$

$0 = 2\pi r (20 - 3\pi r)$

$r = 0 \quad -3\pi r = -20$

$r = \frac{20}{3\pi}$

$V'' = 40\pi - 12\pi r$
 $= 40\pi - 12\pi \left(\frac{20}{3\pi}\right)$

$= -40\pi$
 < 0

$\therefore \text{max.}$

$V = 20\pi \left(\frac{20}{3\pi}\right)^2 - 2\pi^2 \left(\frac{20}{3\pi}\right)^3$

$= \frac{8000}{27\pi}$ cubic units